It is easy to calculate the average value of finitely many numbers $y_1, y_2, \ldots, y_n$:

$$y_{\text{avg}} = \frac{y_1 + y_2 + \cdots + y_n}{n}, \tag{1}$$

but how do we compute the average temperature during a given day if infinitely many temperature readings are possible? The figure below shows that graph of a temperature function $T(t)$, where t is measured in hours and T in °C, as well as a guess at the average temperature, T_{avg}.

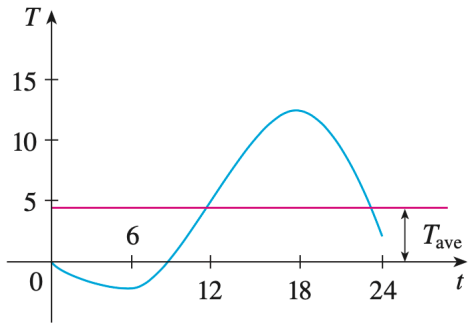


Figure 1: Graph of a temperature function and the average temperature.

More generally, let's try to compute the average value of a continuous function $y = f(x)$, defined on an interval $a \le x \le b$. Start by dividing the interval $[a,b]$ into n subintervals $[x_{i-1}, x_i]$ of equal width $\Delta x = \frac{b-a}{n}$. For each $i = 1, 2, \ldots, n$, choose an arbitrary point x_i^*.

1. Calculate the average of the numbers $f(x_1^*), f(x_2^*), \ldots, f(x_n^*)$ using Equation (1).

2. Substitute $n = \frac{b-a}{\Delta x}$ into your answer to Question 1 and simplify.

3. Recognize your answer to Question 2 as a Riemann sum and take the limit as $n \to \infty$ to define the average value of the function f on $[a,b]$.

Definition. (Average value of a function) If f is a continuous function on $[a,b]$, then the average value of f, f_{avg}, is defined to be

$$\boxed{f_{\text{avg}} := \lim_{n\to\infty} \frac{1}{b-a}\sum_{i=1}^{n} f(x_i^*)\Delta x = \frac{1}{b-a}\int_a^b f(x)dx}.$$

4. Find the average value of the function $f(x) = 1+x^2$ on the interval $[-1,2]$.

If $T(t)$ is the temperature at time t, we might wonder if there is a specific time when the temperature is the same as the average temperature. For the temperature graphed in Figure 1, we see there are two such times - just before noon and just before midnight. In general, is there a number c at which the value of a function f is exactly equal to the average value of the function, that is, $f(c) = f_{\text{avg}}$? The following theorem says that this is true for continuous functions.

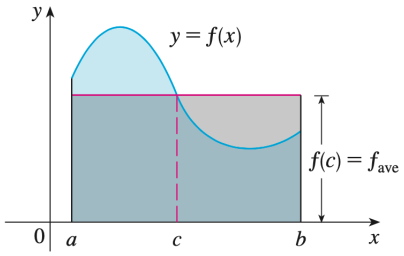
Theorem. (The Mean Value Theorem for Integrals) If f is continuous on $[a,b]$, then there exists a number c in $[a,b]$ such that

$$f(c) = f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)dx,$$

that is,

$$\int_a^b f(x)dx = f(c)(b-a).$$

The geometric interpretation of the Mean Value Theorem for Integrals is that, for *positive* functions f, there is a number c such that the rectangle with base $[a,b]$ and height $f(c)$ has the same area as the region under the graph of f from a to b.



Before proving the Mean Value Theorem for Integrals, let's see some examples.

5. Given $f(x) = 1 + x^2$ on the interval $[-1, 2]$, find all values of c guaranteed by the Mean Value Theorem for Integrals.

6. Show that the average velocity of a car over a time interval $[t_1, t_2]$ (defined as displacement over time) is the same as the average of its velocity function during the trip.

7. Prove the Mean Value Theorem for Integrals by applying the Mean Value Theorem (for derivatives) to the function $F(x) = \int_a^x f(t)dt$.